3DGenerateFlow Competition Poster

> Can be printed from browser or exported to PDF via `docs/generate_poster_pdf.py`.

3DGenerateFlow

One Photo ·One Sentence ·Printable Full-Color 3D / 2.5D Model

Track: Multimodal AI Content Creation Tool Development
Deliverable: Web UI
Platform: AMD Radeon GPU + ROCm Open Software Stack

Why This Project?

- 3D content creation has a high barrier: multi-view capture, modeling, stylization, and print checking are scattered across tools.
- Closed-source APIs are expensive and uncontrollable.
- There is a lack of end-to-end tools for "photo → 3D print."

3DGenerateFlow = Photo + One Sentence + AMD GPU → Full-Color 3D Model / 2.5D Relief

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Core Capabilities

Feature	Description
AI Director Agent	One-sentence planning of style, parameters, and 6-step production flow
Image-to-Image Stylization	Stable Diffusion img2img, up to 1024px, ROCm local inference
Multi-View Synthesis	Zero123 generates front / right / back / left views
Image-to-3D	Hunyuan3D-2 / Hunyuan3D-2mv multi-view 3D generation
2.5D Relief	Depth Anything V2 + height map → colorful GLB + printable STL
Texture Fallback	Front-projection texturing when ROCm cannot compile CUDA extensions
Print Check	Real-time volume, bounding box, and watertight report
Web UI Preview	Embedded Three.js 3D viewer and download

System Architecture

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Web UI (React + R3F)
    ↓
FastAPI + Celery (Eager Mode)
    ↓

  ROCm Adapter   Hunyuan3D-2    3D Director
  · SD img2img     · 2D → 3D      · Agent Plan
  · Zero123        · 2mv Multi-View  · Memory/Chat
  · Depth V2       · Texture Fallback
```

```
        ↓
    Print Check / Export (GLB / STL)
```

Tech Stack

- **Frontend**: React 19 + TypeScript + Tailwind + Vite + R3F
- **Backend**: FastAPI + Celery + SQLAlchemy + SQLite
- **AI**: PyTorch 2.5.1 + ROCm 6.1 + Diffusers + Transformers
- **Models**: Stable Diffusion v1.5, Zero123-xl, Depth Anything V2, Hunyuan3D-2 / 2mv
- **Printing**: Trimesh mesh processing, UV texturing, volume / watertight check

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AMD ROCm Adaptation Highlights

- One-click scripts install ROCm environment + PyTorch for ROCm + Hunyuan3D-2.
- Core inference runs locally on AMD Radeon gfx1100 (48 GB VRAM).
- Fixed Hunyuan3D-2 `to()` returning `None` on ROCm.
- Automatic fallback to front-projection texturing when the texture module's CUDA extensions fail to compile.
- Celery memory backend, no Redis required for demo.

Application Scenarios

- Pet / person 3D figures and mementos
- Fridge magnets, keychains, commemorative coins, relief medals
- Rapid 3D prototyping for products
- Education / makerspace AI + 3D printing teaching

Demo Results

- **2.5D Relief**: 80×80×7 mm, watertight, 30 cm^3 volume, colorful GLB + printable STL.
- **Full-Color 3D**: Multi-view input → 80 mm target height → colorful GLB with Web UI preview.
- **Diverse Styles**: Realistic, cartoon, low-poly, voxel, clay, sketch, lithophane, silhouette, and more.

Open Source & Repository

- **GitHub**: `SuppartWang/3DGenerateFlow`
- **README**: ROCm startup guide, dependency list, Web UI instructions.
- **Docs**: `docs/PROJECT_INTRO_EN.md`, `docs/DEMO_SCRIPT.md`, `docs/POSTER_EN.md`

3DGenerateFlow —Let everyone turn photos into printable 3D memories with AMD GPU.